Project 2 – Acme Analytics Linux Platform

Day 3 · Phase 1A: the service desk – why, how it works, and building svc01

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Project	Acme Analytics – hardened, observable and recoverable Linux platform
Phase	1A – Service desk: GLPI tickets, knowledge base and asset inventory
Host worked on	svc01 (new VM)
Status	Step 1A.1 🟡 – VM created and hardware set; Ubuntu installation next
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> **How to read this report.** 💡 **In plain words** – the idea for any reader. 🔧 **Technical** – the settings and commands. 📸 **Evidence** – the real screenshot. Terms are explained in the glossary.

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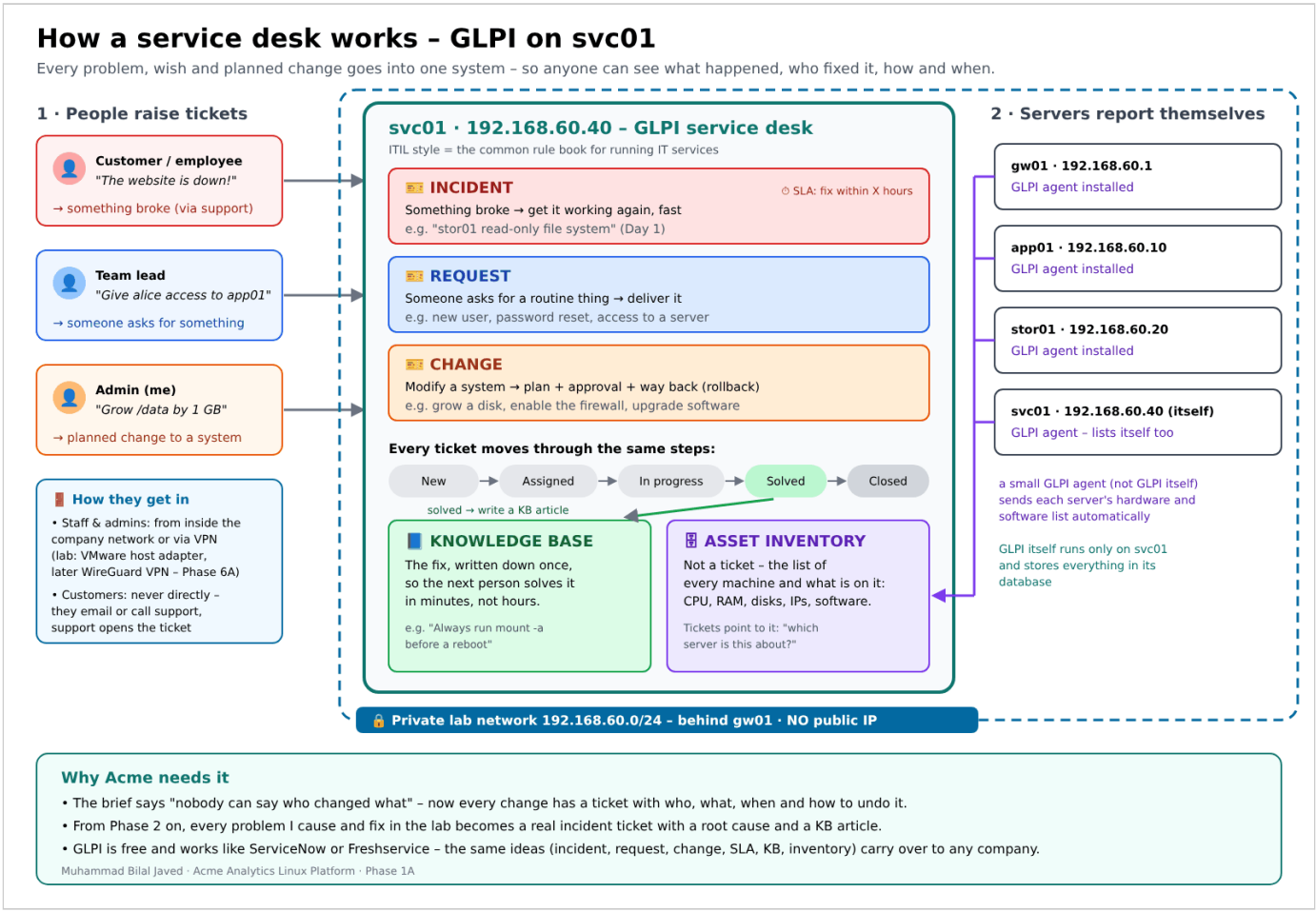
1. The story – why a service desk, and why now

💡 **In plain words:** until now I fixed problems and wrote them down in these reports. A real IT team does not work like that: every problem, every change and every "please give me access" goes into a **ticket system**, so anyone can see *what happened, who fixed it, how, and when*. Phase 1A gives Acme that system.

Where it fits in the Acme story. The brief says: *"nobody can say who changed what"* and *"the first enterprise customer's security questionnaire asks questions the team cannot answer."* The network (Phase 1) now works, so this is the right moment: from Phase 2 on, **every incident I cause and fix becomes a real ticket**, every planned change (e.g. "grow /data by 1 GB") becomes a **change** ticket with a rollback plan, and every server appears in an **asset inventory** automatically.

Ticket type	Meaning	Example in this project
Incident	Something broke	"stor01 read-only file system" (happened on Day 1)
Change	A planned change with a plan and a rollback	"Grow /data by 1 GB", "Enable UFW on stor01"
Request	Someone asks for a routine thing	"Give alice access to app01", a password reset
Knowledge base (KB)	The written fix, so the next person solves it faster	"Always run `mount -a` before a reboot"
Asset inventory	Every server with CPU, RAM, disks, IPs and installed software	gw01, app01, stor01, svc01

📊 **The picture – how a service desk works:**



How a service desk works – GLPI on svc01

Key understanding (in my own words).

> In the IT industry a problem is an **incident**, and handling them is **incident management**. When someone asks for a routine thing – a new user, a password reset, access to a server – it is a **request**. When I modify a system itself – grow a disk, enable a firewall, upgrade software – it is a **change**, which needs a plan, approval and a way back. All three are **tickets** in GLPI, and every ticket moves through the same steps: new → assigned → in progress → solved → closed; a solved fix is written down in the **knowledge base**. The **asset inventory** is not a ticket: it is the list of every machine and what is on it, filled automatically by the GLPI agent on each server, so a ticket can say which server it is about. GLPI follows **ITIL**, the common rule book for running IT services, and is used by many European companies and public bodies.

Who can reach svc01? svc01 has **no public IP** – it sits on the private lab network behind gw01, like app01. Staff and admins reach it from inside the company network (in the lab: my PC through VMware's host adapter; from Phase 6A through the WireGuard VPN). **Customers never reach it directly**: they email or call support, and support opens the ticket for them. A service desk holds the details of every server, so exposing it to the internet would be a security risk.

Why svc01 appears twice in the picture: in the middle it is the **GLPI application** (the help desk); on the right it is the **machine** svc01, which also runs a GLPI agent and lists itself in the inventory. GLPI itself runs only on svc01; the other servers get only the small agent.

Incident or change? "The server is broken, please update it" is an **incident** (something is broken). "The server runs an old version, upgrade it next Tuesday" is a **change** (planned). The SLA – how fast a ticket must be handled – is the IT team's promise, not something the customer sets.

Why GLPI. GLPI is a free, ITIL-style service desk used by many European companies and public bodies. Freshservice or ServiceNow need a paid licence; GLPI teaches the same concepts – incident, change, request, SLA, KB, CMDB – and they carry over to any tool.

Why it matters for my job search. Many of the junior IT / DevOps postings I reviewed ask for ticket systems, SLAs, asset management or knowledge articles. In an interview I can say: *"I ran my lab through GLPI. Every incident I caused has a ticket, a root cause and a KB article, and every server is in the inventory."*

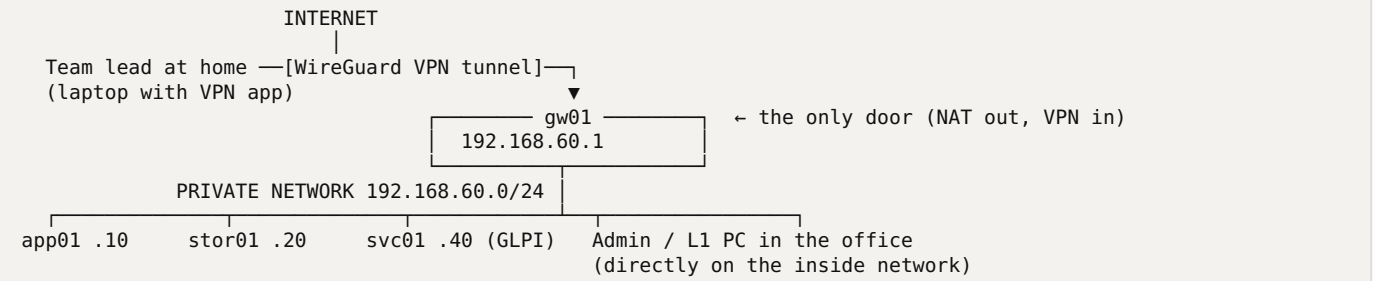
Why a separate server. The service desk must keep working when the servers it tracks are broken – if app01 is down, I still need to open the incident ticket. So it gets its own VM, `svc01`. Later, Project 1 reuses the same svc01 as a second "customer" – one service desk serving two customers is how IT system houses and managed service providers work.

2. svc01 in the lab

VM	Role (plain words)	Role (technical)	IP
gw01	The gatekeeper	NAT, firewall	192.168.60.1
app01	The shop window	App server	192.168.60.10
stor01	The archive room	Storage, backups, logs	192.168.60.20
svc01	**The help desk**	**GLPI tickets, KB, inventory (Docker)**	**192.168.60.40**

2a. How people reach svc01 – and where gw01 comes in

💡 **In plain words:** gw01 is the **only door** between the private network and the internet. People already **inside** (in the office) go straight to svc01. People **outside** (at home) must come in through the door with a **VPN** – like a company badge for remote workers. Strangers and customers never get in.



Who	Where	Path to svc01	Through gw01?
Employee / team lead in the office	On the company network	Browser → `http://192.168.60.40`	❌ already inside
Team lead / admin from home	On the internet	VPN app → encrypted tunnel to gw01 → private network → svc01	✅ via the VPN
Stranger on the internet	Outside, no VPN	Blocked – svc01 has no public IP, gw01 lets in VPN users only	🚫
Customer	Outside	Never connects – emails or calls support; L1 opens the ticket	—
Me in the lab, now	Windows PC	VMware host adapter on VMnet2 → browser → `http://192.168.60.40`	❌
Me in the lab, from Phase 6A	Like the home worker	WireGuard VPN to gw01 → svc01	✅

Lab note: VMware may give the Windows host adapter on VMnet2 the address 192.168.60.1 – gw01's address. Before opening GLPI, check it in the Virtual Network Editor / Windows network settings and use a free address such as 192.168.60.250.

2b. Three examples, start to finish

How the GLPI agent fits in: a small program on each server looks at its own machine once a day (CPU, RAM, disks, IP, software) and sends the report to `http://svc01` – finding svc01 through `/etc/hosts` or DNS. GLPI stores it as an **asset**. The agent creates no tickets and nobody logs in through it. Tickets come from **people**; in each ticket the person picks the affected server ("Item: app01") from the inventory, which links the ticket to the machine.

Thing	Question it answers	Used by
`/etc/hosts` / DNS	"What is stor01's IP?"	Machines, to connect
Asset inventory	"What is app01, what does it have?"	People, to know and plan
Ticket	"What is wrong or needed, and who did what?"	People, to work

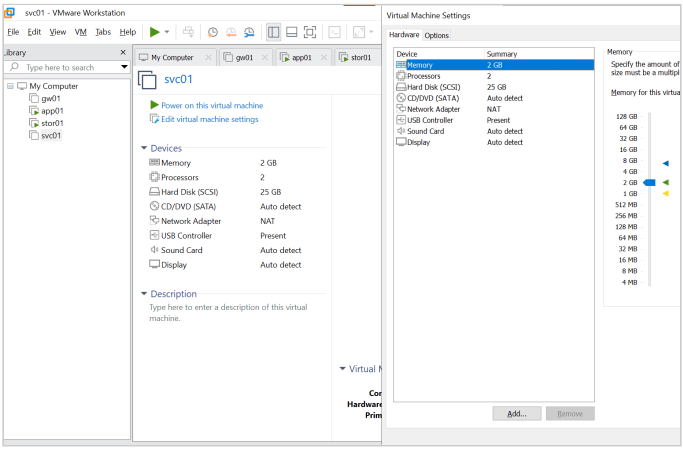
1. **Alice (employee) → Incident.** "I can't open the shared folder on app01." Item: app01. L1 checks (app01 pings, folder hangs) → escalates to **Linux Admins (L2)** with notes → L2 finds the firewall blocking NFS on stor01, fixes it, writes the root cause → Solved → KB article → alice confirms → Closed.
2. **Team lead → Request.** "New developer carol starts tomorrow; account on app01 in group devs." L1 runs `useradd -m -s /bin/bash -G devs carol` and `passwd carol` → Solved → Closed. When carol leaves: a new request → `usermod -L carol`.
3. **Admin → Change.** "/data on stor01 is 90 % full." The inventory shows 4 GB free in the storage pool → change ticket "Grow /data by 1 GB" with plan (`lvextend -r -L +1G`), risk, rollback (LVM snapshot first) and approval → done → `df -h` checked → Closed. The next agent run shows the new size automatically.

In the lab I play every role with separate GLPI accounts: `alice`/`bob` as requesters, `tech.bilal` in the groups **Service Desk (L1)** and **Linux Admins (L2)**. Escalation = reassigning the ticket to the higher group with notes; the history stays with the ticket.

3. VM settings

Setting	Value	Why
OS	Ubuntu Server 24.04 LTS	Current LTS; GLPI runs in Docker on it
RAM	2 GB	GLPI + its database in Docker
CPU	2 vCPU	Smooth web interface; 1 would also work
Disk	25 GB	Database, ticket attachments, Docker images
Network	**One adapter, Custom → VMnet2**	Lab network only, like app01 and stor01 – internet only through gw01's NAT
CD/DVD	Ubuntu Server ISO, connect at power on	Installation medium
Sound card, USB	Optional – can be removed	A server does not need them
Hostname / user	`svc01` / `bilal_004`	Same naming pattern as the other servers
IP (set in the installer)	`192.168.60.40/24`, gateway `192.168.60.1`, DNS `8.8.8.8`	VMnet2 has no DHCP, so the address is set by hand
Installer option	✅ Install OpenSSH server	So gw01 can manage it over SSH with keys

📸 Evidence – VMware's default settings: 2 GB, 2 CPUs, 25 GB, but the network adapter is on NAT:

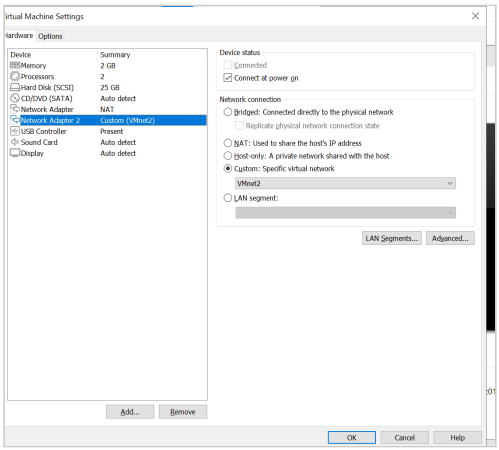


svc01 default settings with NAT

Issue: network adapter on NAT

Seen	VMware creates a new VM with its adapter on **NAT** (direct internet). Then a second adapter was added on VMnet2, so the VM had two adapters.
Why it matters	With NAT, svc01 would bypass gw01 – the gatekeeper design from Phase 1 would be broken, and the NAT adapter would have to be unplugged and netplan fixed after the install (the same trouble app01 had on Day 1).
Fix	Keep one adapter on **Custom → VMnet2**; **remove** the NAT adapter.
Lesson	Check every new VM's network before the first boot – the default is not the design.

📸 Evidence – Network Adapter 2 on Custom (VMnet2), connect at power on:



adapter on VMnet2

4. Next steps

1. Remove the NAT adapter, attach the Ubuntu ISO, power on (gw01 running).
2. Install Ubuntu with the manual network settings above; tick OpenSSH.
3. Test: `ping -c 2 192.168.60.1` and `ping -c 2 8.8.8.8` (internet through gw01).
4. Add `192.168.60.40 svc01` to `/etc/hosts` on all servers; copy bilal_001's SSH key from gw01.
5. Step 1A.2: install Docker and start GLPI.